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Healthy brain aging assessed with [^{18}F]FDG and [^{11}C]UCB-J PET

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ABSTRACT

Background: The average human lifespan has increased dramatically over the past century. However, molecular and physiological alterations of the healthy brain during aging remain incompletely understood. Generalized synaptic restructuring may contribute to healthy aging and the reduced metabolism observed in the aged brain. The aim of this study was to assess healthy brain aging using [^{18}F]FDG as a measure of cerebral glucose consumption and [^{11}C]UCB-J PET as an indicator of synaptic density.

Method: Using in vivo PET imaging and the novel synaptic-vesicle-glycoprotein 2A (SV2A) radioligand [^{11}C]UCB-J alongside with the fluorodeoxyglucose radioligand [^{18}F]FDG, we obtained SUVR-1 values for 14 pre-defined volume-of-interest brain regions defined on MRI T1 scans. Regional differences in relative [^{18}F]FDG and [^{11}C]UCB-J uptake were investigated using a voxel-wise approach. Finally, correlations between [^{11}C]UCB-J, [^{18}F]FDG PET, and age were examined.

Results: We found widespread cortical reduction of synaptic density in a cohort of older HC subjects ($N = 15$) compared with young HC subjects ($N = 11$). However, no reduction persisted after partial volume correction and corrections for multiple comparison. Our study confirms previously reported synaptic stability during aging. Regional differences in relative [^{18}F]FDG and [^{11}C]UCB-J uptake were observed with up to 20 % higher [^{11}C]UCB-J uptake in the amygdala and temporal lobe and up to 34 % higher glucose metabolism in thalamus, striatum, occipital, parietal and frontal cortex.

Conclusion: In vivo PET using [^{11}C]UCB-J does not support declining synaptic density levels during aging. Thus, loss of synaptic density may be unrelated to aging and does not seem to be a sufficient explanation for the recognized reduction in brain metabolism during aging. Our study also demonstrates that the relationship between glucose consumption and synaptic density is not uniform throughout the human brain with implications for our understanding of neuroenergetics.

1. Introduction

The average human lifespan has increased dramatically over the past century. However, underlying molecular mechanisms of the healthy brain during aging remain incompletely understood. During the latter decades, advanced brain imaging techniques have provided novel insights. Several authors have investigated the influence of age on brain metabolism using positron emission tomography (PET) imaging with fluorine-18-fluorodeoxyglucose ([^{18}F]FDG PET) in vivo in healthy subjects. Such studies demonstrate progressively reduced metabolism in the aging brain, ranging from 2 to 10 % per decade [1–4], mainly in frontal and parietal brain regions [3–7]. This reduced glucose metabolism can to some extent be explained by cerebral atrophy including loss of grey matter (GM) density e.g. synaptic loss, cortical thinning, and decreases in white matter (WM) volume, leading to an overall loss of total brain matter [4,8–11]. However, previous studies indicate that atrophy alone is not sufficient to explain the reduced metabolism with age [3,12] and previous pathological studies confirm that direct loss of

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neurons is limited during aging [13]. Additionally, one comprehensive stereology study only found a rather small change of 10 % in the number of neocortical neurons as a function of age [14]. Thus, the observed reduced metabolism and cerebral atrophy in the aging brain might be explained by generalized synaptic pruning and/or degeneration.

Several postmortem studies suggest synaptic density to be stable during aging [15–17]. However, one postmortem study suggests a decrease of presynaptic terminals in the frontal lobe measured by the gold standard synaptic marker synaptophysin [18]. However, this has not been confirmed by in vivo imaging techniques. The recently developed PET tracer [^{11}C]UCB-J can quantify the density of the presynaptic vesicle protein SV2A [19]. This protein is ubiquitously present in all presynaptic terminals and thus considered a proxy of synaptic density [19–21].

Michiels and colleagues recently published the first study looking at synaptic density in vivo in healthy aging [22]. They observed stable [^{11}C]UCB-J binding with healthy brain aging in vivo, indicating that synaptic density is preserved [22]. They did not incorporate [^{18}F]FDG PET scans as a measure of brain metabolism in that study. Thus, the relationship between synaptic density and glucose metabolism in healthy aging remains unexplored.

Here, we used [^{11}C]UCB-J to study cerebral synaptic integrity and [^{18}F]FDG PET to study cerebral metabolism in 26 healthy subjects aged 22–82 years. We hypothesized that loss of synaptic density is related to the decline in cerebral metabolism during normal aging. Furthermore, we investigated if the regional glucose uptake normalized to synaptic density is changing in specific brain regions because of aging.

2. Materials and methods

2.1. Study design and participants

A total of 26 healthy subject were recruited – 11 young subjects and 15 older subjects. Clinical and demographic data are presented in Table 1. All subjects were recruited through advertisement in local newspapers.

The study was conducted between March 2019 and December 2020. Study subjects of specific age groups were included (20–30 or 65–85 years of age). (Montreal Cognitive Assessment >26). Exclusion criteria were Montreal Cognitive Assessment score < 26, clinical depression or clinically significant symptoms of depression (Geriatric Depression Scale (GDS-15) score > 6), schizophrenia, schizoaffective disorder, bipolar disorder or any history of electro-convulsive therapy (ECT), past concussive head injury, stroke, major systemic diseases or medical conditions, significantly impaired hearing or sight that could influence neuropsychological assessment, significant white-matter lesions observed on the T1 and T2 MRI scan, current or previous cancer, and medications affecting SV2A binding (e.g. levetiracetam).

This study was approved by the Central Denmark Region Committees on Biomedical Research Ethics (case nr. 1-10-72-160-18) and was conducted in accordance with the Declaration of Helsinki. All subjects provided informed written consent.

Table 1
Demographic and clinical characteristics of study subjects.

	HC young	HC old	P
Sample size, n	11	15	
Age	24.8 (±1.2)	72.7 (±4.2)	<0.001
Sex, male/female	7/4	8/7	0.599
MoCA	29.4 (±0.77)	28.4 (±1.14)	0.057
Z-score memory	0.12 (±0.60)	−0.05 (±0.77)	
Z-score attention	0.17 (±0.23)	0.07 (±0.76)	
Z-score executive function	0.20 (±0.64)	0.12 (±0.54)	
Z-score language	0.03 (±0.31)	−0.24 (±0.73)	
Z-score visuospatial function	0.18 (±0.70)	0.14 (±0.90)	
Z-Score Average	0.14	0.01	

Data shown as mean ± SD.

2.2. [^{18}F]FDG, [^{11}C]UCB-J PET and MRI imaging

A total of 26 healthy subjects underwent PET imaging with both [^{11}C]UCB-J and [^{18}F]FDG on two separate scan days, and had an additional structural MRI on a third scan day.

MRI scans were performed on a 3 T Siemens SKYRA magnetic resonance system. The T1-weighted MPRAGE sequences were used for co-registration with [^{11} C]UCB-J and [^{18}F]FDG PET and for volume of interest definition. T1 and T2 images were used to exclude cases with anatomical abnormalities and major pathology.

[^{11}C]UCB-J scans and tracer radiosynthesis were performed as previously described [21,23,24]. An intravenous bolus of 500 MBq (±10 %) [^{11}C]UCB-J was administered 60 min prior to PET in all subjects. A 30-min dynamic PET scan was acquired on the ECAT Siemens High-Resolution Research Tomograph (Siemens/CTI, Knoxville, TN, USA). Prior to [^{11}C]UCB-J, a 6-minute transmission scan was acquired for attenuation correction.

All subjects had [^{18}F]FDG PET on the Siemens Biovision PET/CT system. Intra-venous injection with 120 MBq (±10 %) [^{18}F]FDG was administered, which was followed by a 20-minute PET scan acquired 30 min after injection. Prior to [^{18}F]FDG PET, a low-dose attenuation CT scan was acquired to correct for tissue attenuation effects. All subjects fasted for at least 4 h before [^{18}F]FDG PET. Subjects were positioned comfortably in a quiet, dimly lit room several minutes before FDG administration and afterwards remained there for at least 20 min. They were instructed not to speak, read or be otherwise active.

We processed PET images with PMOD™ 4.0 software using relevant toolboxes including Gaussian kernel smoothing with an FWHM (full width at half maximum) of 8 mm, normalization of PET and MRI to Montreal Neurological Institute (MNI) space, rigid matching of each subject's PET to their anatomical T1 MRI, MRI segmentation into grey and white matter images and volume-of-interest (VOI)-based (Geometric Transfer Metrix) partial volume correction (PVC) [25,26]. All PET analyses were done with PVC for the VOI-based approach. PVC was considered mandatory for correct interpretation of PET data since an aging effect on grey matter was expected [27]. A considerable difference was observed after implementing PVC on our data; hence, all results show representative values corrected for PVC effects.

Parametric [^{11}C]UCB-J uptake value ratio (SUVR) images were generated using the centrum semiovale (CSO) as a reference region, as previously validated in several studies [28–31]. The centrum semiovale was delineated by the use of the white matter regions and then minimizing it in PMOD using the isocontour tool (PMOD4.0, PMOD technologies, Zurich, Switzerland) to include only voxels with a high probability of being part of the centrum semiovale, resulting in a restrictive oblong reference region in the center of the CSO. This region was then added to the standard Hammers atlas and applied on all subjects. [^{18}F]FDG SUVR images were generated using two different reference regions; CSO and grey matter (GM) [32]. Subsequently, all other regions were defined from the built-in Hammers N30R83 atlas, and the correct VOI placement was ensured by overlaying the automatically generated VOIs on the MRI scan in atlas space. To reduce dimensionality and avoid type II errors, some of the standard bilateral VOIs were merged into 14 larger composite regions comprising both left and right hemispheres. After careful inspection, it was concluded that no other manual corrections to the VOI placement were necessary for all VOIs. Thus, we assessed [^{18}F]FDG and [^{11}C]UCB-J uptake in these 14 brain regions relevant to healthy brain aging and similar to a previously published study [32]: ACCx: anterior cingulate cortex, CBL: cerebellum, FCx: frontal cortex, ICx: insular cortex, LTCx: lateral temporal cortex, mTCx: medial temporal cortex, OCx: occipital cortex, PCCx: posterior cingulate cortex, PCx: parietal cortex, Str: striatum, Thal: thalamus.

In addition to CSO normalization, we also normalized to GM to generate another set of parametric SUVR images. The whole brain GM

normalization was generated using the individual GM mask obtained after PMOD normalization and segmentation with the use of T1-weighted MRI images. The GM normalized parametric image output was used for voxel-based analyses and creation of relative [^{18}F]FDG and [^{11}C]UCBJ intensity maps.

2.3. Neuropsychological assessment

All subjects completed a battery of neuropsychological tests, as previously described [23]. See online supplementary material for details about specific neuropsychological tests. The average Z score for each cognitive domain was calculated for both of the healthy groups (Table 1). Z-scores were based on age-matched published norms [33,34]. No difference was observed between the young and old group with respect to their performance on neuropsychological tests.

2.4. Statistical analysis

General statistics were conducted in SPM12 on a MATLAB 2019a platform, GraphPad PRISM (version 6), and RStudio (v1.1.463. RStudio, Inc., Boston, MA). Normal distribution of data was assessed with Shapiro-Wilk tests, Q-Q plots and box plots prior to any statistical analysis. Grubbs' test was used to identify statistical outliers. Group comparisons of [^{11}C]UCB-J and [^{18}F]FDG binding ratios were assessed with a *t*-test for pairwise comparisons of means with equal variance. Significance level was set at a threshold of $P < 0.05$. Bonferroni correction was used for multiple comparisons. For voxel level between group comparisons, paired *t*-tests were performed in SPM12 between the relative [^{18}F]FDG and [^{11}C]UCB-J parametric maps (voxel and cluster-level FWE $P < 0.05$ and cluster extent threshold $k = 300$ voxels). The 300 voxels correspond to a volume of 300 mm^3 (voxel size $1 \times 1 \times 1$ mm). Voxels surviving that threshold were overlaid on a rendered brain from SPM12. For each VOI, a simple linear regression between [^{11}C]UCB-J, [^{18}F]FDG SUVR, and age was calculated.

3. Results

3.1. Demographics

Demographic and clinical characteristic are summarized in Table 1. No significant between-group differences in sex distribution, MoCA score, or neuropsychological z-scores were observed ($P > 0.05$). No subjects had signs of cognitive impairment as assessed by MoCA.

3.2. VOI analysis

As demonstrated in Fig. 1A and C, the aged group on average showed reduced overall [^{11}C]UCB-J and [^{18}F]FDG uptake compared to the young subjects after CSO intensity normalization. VOI-based [^{11}C]UCB-J SUVR-1 and [^{18}F]FDG SUVR binding group differences using a CSO

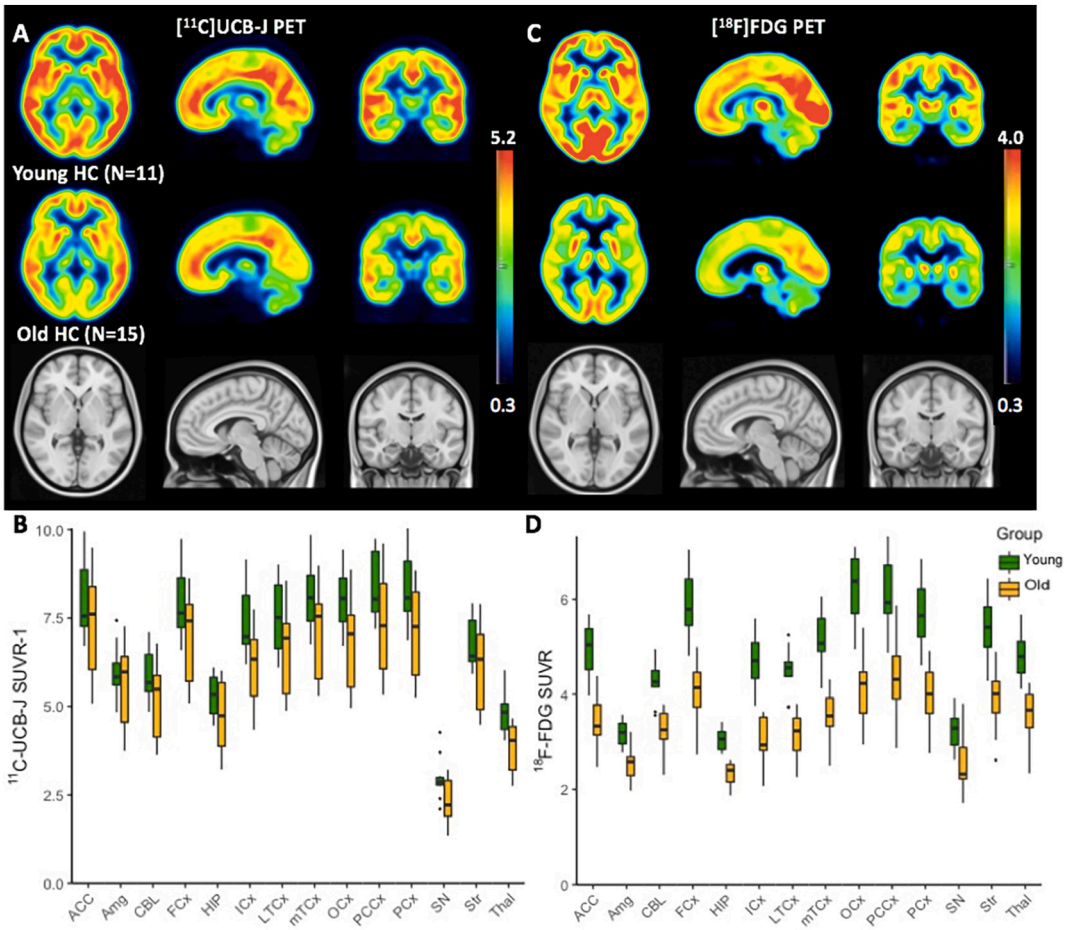


Fig. 1. (a) Average group [^{11}C]UCB-J PET image of the brain shown for visual comparison. Colour scale is SUVR values. (b) Box-and-whisker plot of PVC regional [^{11}C]UCB-J binding for the young HC group in green and the old HC group (yellow). (c) Average group [^{18}F]FDG PET image of the brain shown for visual comparison. Colour scale is SUVR values. (d) Box-and-whisker plot of PVC regional [^{18}F]FDG binding for the young group in green and old group (yellow). ACCx: anterior cingulate cortex, CBL: cerebellum, FCx: frontal cortex, ICx: insular cortex, LTCx: lateral temporal cortex, mTCx: medial temporal cortex, OCx: occipital cortex, PCCx: posterior cingulate cortex, PCx: parietal cortex, Str: striatum, Thal: thalamus. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

reference are shown in a box and whisker plot in Fig. 1B and D. Reduced [^{11}C]UCB-J SUVR-1 values were observed in 10 of 14 regions, with only Hip, AMG, Str, and ACC not showing significant reductions. However, no significant reduction persisted after PVC and Bonferroni correction for multiple comparisons (Table S2). Significantly reduced [^{18}F]FDG SUVR values in the aged group were observed in all 14 regions, and all reductions persisted after PVC and Bonferroni correction for multiple comparison (Table S2).

3.3. Voxel-wise comparison

The universal reduction observed in Fig. 1A and C was quantitatively visualized by voxel-wise comparison in SPM12. For [^{11}C]UCB-J SUVR-1 values, only small sporadic voxel clusters showed significant differences between the young and older subjects (Fig. 2A). However, in accordance with the VOI analyses, the voxel-wise comparison showed a comprehensive and almost universal reduction of [^{18}F]FDG SUVR values in the older subjects compared with the young group (Fig. 2B).

3.4. Parametric maps of relative [^{18}F]FDG vs. [^{11}C]UCB-J uptake differences

Fig. 3A represents a voxel-wise parametric difference map as percentages for the relative distribution of [^{18}F]FDG and [^{11}C]UCB-J, normalized to the individual average whole brain GM uptake.

In accordance with a previously published study [32], the figure shows higher relative [^{11}C]UCB-J symmetrical uptake (red and yellow colours) in the lateral and medial temporal lobe, anterior cingulate cortex, brain stem, insula cortex, and the highest difference in the amygdala. Opposite, blue colours represent significant symmetric voxels with higher relative [^{18}F]FDG uptake in the frontal cortex, posterior cingulate cortex, cerebellum, occipital cortex, striatum, and thalamus. For quantitative visualization, VOI-based [^{11}C]UCB-J SUVR-1 and [^{18}F]FDG SUVR relative binding group comparisons are shown in a histogram in Fig. 3B. This plot confirms a maximum relative [^{18}F]FDG uptake< [^{11}C]UCB-J uptake in the AMY (20 %), LTCx (17 %), ACCx (14 %), mTCx (13 %) and ICx (10 %); and a maximum relative [^{11}C]UCB-J uptake< [^{18}F]FDG uptake in the Thal (34 %), Str (11 %), OCx (9 %), CBL (8 %), PCCx (4 %) and FCx (4 %). Corresponding [^{18}F]FDG vs. [^{11}C]UCB-J difference maps and VOI parameters for the young and old group separately are shown in online supplementary Fig. S1, and Table S1. Medial and axial projections are shown in online supplementary Fig. S2.

4. Discussion

In the present study, we report widespread universal reduction of [^{11}C]UCB-J binding, suggestive of a declining number of synapses in healthy brain aging. However, no significant reduction of synaptic density persisted after PVC and correction for multiple comparisons (Fig. 1B). This finding could signify that the well documented overall loss of grey matter volume during aging is caused by declining numbers of synapses, but that the density of synapses per volume grey matter remains constant with age. This notion is in agreement with previous pathological studies showing that neuron loss is limited during aging [13,14]. Our results (Fig. 1) confirm the previous observations by Michiels et al. [22], who also showed that PVC-corrected cerebral [^{11}C]UCB-J PET uptake does not decline with age [22].

We saw widespread, significant reduction of [^{18}F]FDG during aging in both the VOI- and voxel-based analysis (Figs. 1C and D, 2B). We had expected to find the age effect to be most pronounced in frontotemporal areas as shown by several previous studies. However, in our dataset, we observed a more homogeneous cortical age effect on our CSO normalized [^{18}F]FDG SUVR. This could be due to our limited sample size of 11 young versus 15 old HC. In addition, the definition of the reference region CSO and its proximity to other tissue classes (grey matter and CSF) can influence and potentially bias SUVR data. However, CSO was defined with caution and manually checked and corrected on every axial slice of each subject. Further, when using the same reference region on our [^{11}C]UCB-J images from the same individuals, we found no significant age-reduction after PVC and correction for multiple comparisons. This finding was in compliance with previously published data [22].

The widespread reduction of [^{18}F]FDG during aging in both the VOI- and voxel-based analysis (Figs. 1 C and D, 2B) suggest that the magnitude of the reduced [^{18}F]FDG uptake with age seemed to be greater than

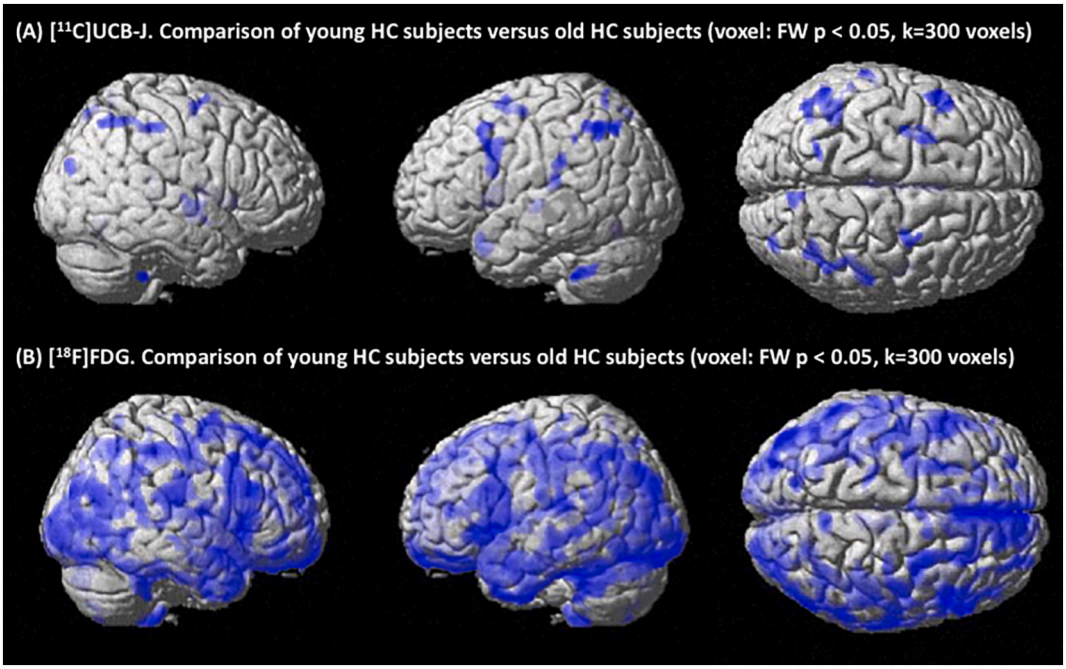


Fig. 2. Reduced PET signal in older HC subjects. (a) Comparison of young HC ($N = 11$) and old HC ($N = 15$) [^{11}C]UCB-J SUVR (voxel and cluster: FWE $P < 0.05$ with cluster thresholding, k = 300 voxels). (b) Comparison of young HC (N = 11) and old HC (N = 15) [^{18}F]FDG SUVR (voxel and cluster: FWE P < 0.05 with cluster thresholding, k = 300 voxels).

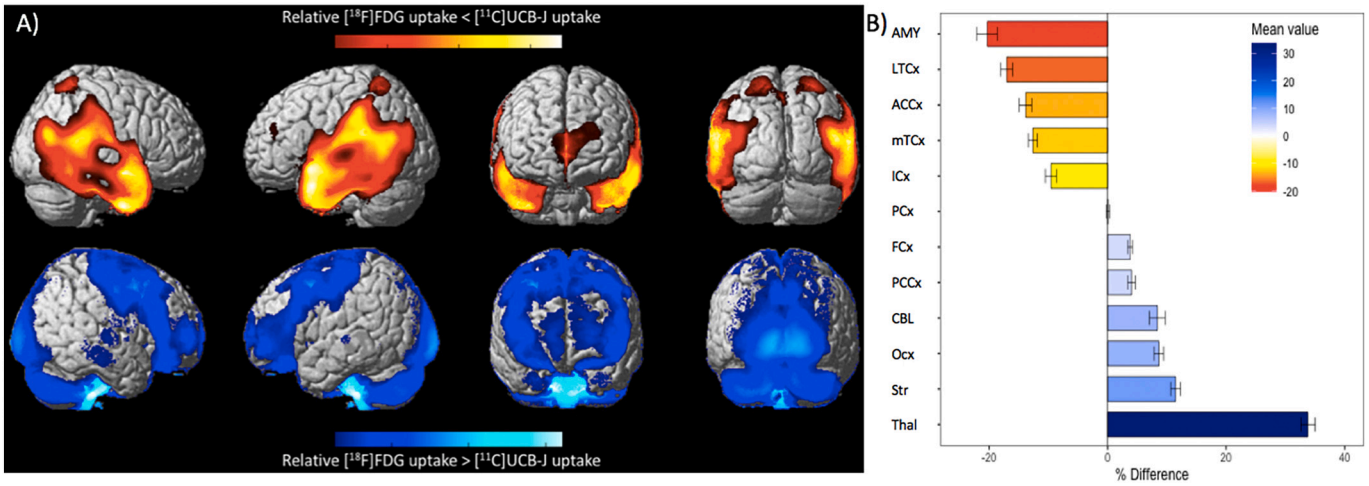


Fig. 3. (a) SPM paired t-statistic parametric maps of relative [^{18}F]FDG uptake<[11C]UCB-J uptake (upper row) and [^{18}F]FDG > [^{11}C]UCB-J (bottom row). Only significant clusters with FWE < 0.05 are shown. Colour bars above and below the Figure represent the t-value. (b) Percentage difference of [^{18}F]FDG over [^{11}C]UCB-J for the main VOIs. All HC subject ($N = 26$). Error bars represent the standard deviation between subjects. ACCx: anterior cingulate cortex; CBL: cerebellum; FCx: frontal cortex; ICx: insular cortex; LTCx: lateral temporal cortex; mTCx: medial temporal cortex; OCx: occipital cortex; PCCx: posterior cingulate cortex; PCx: parietal cortex; Str: striatum; Thal: thalamus.

the decline in [^{11}C]UCB-J binding. In short, even if synaptic loss is responsible for a significant part of grey matter loss with increasing age, the concomitant decrease in [^{18}F]FDG uptake seems to exceed that loss. Thus, it is possible that each synapse uses progressively less energy as we age. However, many other factors are likely to also affect this relationship. For instance, a recent study demonstrated that age-dependent decrease in cerebral glucose metabolism appeared to be related to microstructural WM changes, indicating that parallel and interconnected WM changes contribute to decreased energy consumption with age [4]. It has also been reported that the relative number of subpopulations of synapses (GABA-ergic and glutamatergic) may show regional changes with aging in limbic and prefrontal regions, which could alter regional energy consumption despite an unaltered total number of synapses per volume of grey matter [35,36].

Regional differences in relative [^{18}F]FDG vs. [^{11}C]UCB-J uptake was observed, with up to 20 % higher [^{11}C]UCB-J uptake in the amygdala and temporal lobe and up to 34 % higher glucose metabolism in thalamus, striatum, occipital, parietal, and frontal cortex (Fig. 3A and B). In short, our data suggests some synapses are on average more “hungry” than others. Our findings are in broad agreement with a recently published study of young healthy female subjects [32]. The authors suggested that the pattern of higher relative [^{18}F]FDG uptake is similar to that seen in areas characterized by a higher amount of aerobic glycolysis at rest, as previously observed by Vaishnavi et al. [37]. This is an intriguing hypothesis, which should be addressed in future studies, and which could improve our understanding not only about fundamental aspects of normal brain function, but also how neuroenergetic parameters may be perturbed and altered by diseases affecting synaptic function and integrity such as neurodegenerative diseases [38,39].

The only previous study included only young, healthy female subjects alone [32]. In contrast, we also aimed to study whether the regional energy consumption per synapse is altered as a consequence of aging (Fig. S1C), since we predicted that we would observe a shift in certain brain regions, some expressing more and some less ‘hungry’ synapses during aging. Interestingly, we observed a shift in glucose consumption per synapse in the frontal lobe from an average of 5 % in the young group to 2 % in the old group (Table S1 C, P 0.001), which suggests lowered glucose consumption by frontal synapses with increasing age. However, the modest sample size of the present study precludes robust conclusions about this finding, which need to be reproduced in future cohorts.

The present study has a number of limitations. First, our sample size was modest with only 11 young healthy subjects and 15 older healthy subjects. Consequently, our study had limited statistical power, which we accommodated by limiting the number of analyses and controlling for multiple comparisons. It is possible that synaptic density actually do show a low magnitude reduction with healthy aging, but our inability to detect this was caused by limited statistical power. Additionally, the small predominance of male sex (although not significant, $p = 0.599$) in the younger group (7/4) compared to the older group (8/7) could possibility have influenced our results.

Second, in this study we mainly used the CSO as reference tissue. Several studies have confirmed the CSO as a valid reference region for analysing [^{11}C]UCB-J data [28–31,39,40]. Nevertheless, white matter is a different tissue class than grey matter and therefore theoretically a suboptimal reference region. In addition, the degree of tracer uptake in CSO (WM) could be influenced by WM pathology, especially in our elderly subjects. A recently published study evaluated both absolute measures and normalized measures to reference regions for differentiation between Alzheimer's disease and control subjects [41]. Likewise, the present results should preferably be reproduced with arterial blood sampling and full kinetic modeling (absolute measures) in addition to normalized measures.

Third, partial volume effect (PVE) is identified as a limitation of most PET studies. In the present study, we performed partial volume correction (PVC) on our PET data, but it is known that brain PET quantification is still often affected after PVC [25]. We cannot exclude that atrophy may have influenced the results, even after PVC. PVE is mainly explained by a net spill-out for high-uptake regions, which leads to an underestimation of the measured tracer uptake in addition to a net spill-in to low-uptake regions, which causes overestimation of the measured tracer uptake [27]. Furthermore, it is important to consider that the possible magnitude of PVE also depends on the size of each region, i.e., smaller regions typically suffer greater PVE than larger regions. In the present study, we confirmed that partial volume effects had a measurable influence on our results (Table S2).

Another limitation is related to the new [^{11}C]UCB-J PET tracer. SV2A imaging is for now considered a proxy of synaptic density and in our results, we assume that the number of SV2A vesicles per synapse is not influenced by age. However, this has to our knowledge not been formally shown in post mortem brain tissue comparing young and old subjects. Thus, the present findings should be interpreted with caution.

It is well known that brain metabolism decreases during aging, especially in frontal and parietal brain areas. In conclusion, the present

[^{11}C]UCB-J data suggest that a corresponding regional decrease in synaptic density is not seen in these brain areas. Thus, loss of synaptic density does not seem to be a major contributor to the recognized reduction in brain metabolism during aging. In both young and old subjects, we found a heterogeneous relationship between regional glucose uptake and the corresponding synaptic density in those regions. This suggests that some cerebral regions have a higher glucose consumption and therefore energy turnover per synapse than others. Interestingly, we found evidence that this relationship may be slightly changed with aging, particularly in the frontal lobes. Future in vivo PET studies exploring the relation between synaptic density and cerebral glucose metabolism are needed to verify our findings, and to study whether the relationship between synaptic density and glucose metabolism is altered in different brain disorders.

Author contribution statement

K.B.A, P.B., A.K.H., D.J.B., M.F.D. designed or organized the study. K.B.A, K.K., A.C.S., performed research. K.B.A analyzed data. K.B.A, P.B. wrote the paper. All authors revised the manuscript.

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Declaration of competing interest

Nothing to disclose.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.nucmedbio.2022.06.007.

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